Learning Source Reliability in Informed Neural Processes

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Abstract

Knowledge-conditioned meta-learning faces a fundamental tension: structured knowledge can sharpen predictions under correct specification but degrade them under misspecification. We show that this tension is architectural, not a matter of training. The Informed Neural Process (INP) fuses data and knowledge via additive combination, creating a contamination gap that no amount of data can overcome. We propose the Bayesian Informed Neural Process (BINP), which replaces additive fusion with precision-weighted Bayesian combination: two separate encoders produce independent Gaussian beliefs, combined so the more precise source dominates. This provides a useful property: wrong knowledge is asymptotically overridden while correct knowledge always reduces posterior variance. The architecture alone improves informativeness but cannot detect misspecification without calibration signal. A two-phase corruption training protocol provides this signal, teaching a learned disagreement mechanism when to widen knowledge uncertainty. Together, the two mechanisms yield substantially lower error than identically trained baselines under severe misspecification, while matching performance for correct knowledge. We validate on both synthetic and real-world temperature data.

1 Introduction

Knowledge-conditioned meta-learning faces a tension that additive fusion cannot resolve. Structured domain knowledge, such as pharmacokinetic parameters, physical constants, and expert annotations, is informative when it matches the data-generating process but harmful when it does not. Any model that fuses knowledge unconditionally must choose between exploiting correct knowledge and surviving wrong knowledge. We show that this is not a matter of training but of architecture.

The Informed Neural Process (INP; Kobalczyk and van der Schaar, 2025) integrates knowledge via additive fusion, $e = \mathrm{ReLU}(R + k)$. This treats both sources as equally reliable with no mechanism to modulate knowledge influence based on data. We show that the resulting contamination gap is bounded away from zero regardless of context size (§4). Knowledge dropout, the INP's robustness mechanism, trains the model to function when knowledge is *absent* ($k = 0$), not when it is *wrong* ($k = k_{\mathrm{w}}$). The informativeness-robustness tension is unresolvable under additive fusion.

We propose the **Bayesian Informed Neural Process (BINP)**, which resolves this tension through two complementary mechanisms. The **architecture** replaces additive fusion with precision-weighted Bayesian combination: two separate encoders produce independent Gaussian beliefs over the latent variable z, combined via the Gaussian product so the more precise source dominates. This provides structural properties that no training strategy on additive fusion can match. The **corruption training protocol** calibrates this architecture for finite-sample reliability detection: a two-phase schedule exposes the model to corrupted knowledge, teaching a learned disagreement mechanism when to widen σ_K. The architecture provides the structural capacity for source-specific weighting; corruption training calibrates it.

Contributions.

- **The informativeness-robustness tension** in knowledge-conditioned meta-learning. We formalize this as the contamination gap (§4): additive fusion creates a constant lower bound on knowledge influence that no amount of data can overcome.
- **Precision-weighted Bayesian combination.** The BINP architecture resolves the tension structurally. Wrong knowledge is asymptotically overridden (Proposition 1), while correct knowledge always reduces posterior variance.
- **Corruption training as calibration.** Without exposure to wrong knowledge, the architecture improves informativeness but cannot detect misspecification. Two-phase corruption training separates representation learning from reliability calibration.
- **Architecture and training are complementary.** A clean BINP outperforms the INP for correct knowledge. Corruption training on the INP improves robustness at moderate perturbation. Only their combination dominates under severe misspecification.
- **Real-world validation.** On a real-world temperature dataset with noisy forecasts, the BINP_c degrades gracefully while the identically trained INP_c degrades substantially, confirming per-sample precision weighting beyond synthetic data.

2 Background and Limitation

2.1 The Informed Neural Process

The INP (Kobalczyk and van der Schaar, 2025) extends the Neural Process family (Garnelo et al., 2018) with structured domain knowledge K. Given context data $D_C = \{(x_i, y_i)\}_{i=1}^n$ and knowledge K about the task function f, the INP models the predictive posterior via a latent variable z:

$$p_\theta(y \mid x, D_C, K) = \int p_{\theta_d}(y \mid x, z)\, q_{\theta_e}(z \mid R, k)\, dz,$$

where $R = h_{\theta_{e,D}}(D_C)$ is the aggregated data representation (mean pooling over encoded context pairs) and $k = h_{\theta_{e,K}}(K)$ is the knowledge embedding (set encoder). Training maximizes the ELBO.

The key assumption is *D2* (knowledge truthfulness): K is consistent with the true data-generating process f. Under D2 and conditional independence $D \perp K \mid f$, the original paper's Theorem 1 shows that knowledge helps in expectation.

2.2 Knowledge Fusion

The INP fuses data and knowledge at the latent encoder input: $e = \text{ReLU}(R + k)$. Both sources enter with equal, unconditional weight. Knowledge dropout ($p = 0.3$) sets $k = 0$ for a fraction of training batches, teaching the model to function when knowledge is absent. At test time, however, wrong knowledge is always present. Dropout cannot suppress k_w.

3 Method

3.1 Two-Pathway Architecture

The BINP replaces the INP's single latent encoder with two separate pathways producing independent Gaussian beliefs over the latent variable z. Each pathway outputs a mean and variance, parameterized via bounded softplus ($\sigma = 0.01 + 0.99 \cdot \text{softplus}(\tilde{\sigma})$):

$$\begin{aligned} \text{Data:} \quad & q_D(z \mid R) = \mathcal{N}(\mu_D,\, \sigma_D^2 I), && (\mu_D, \sigma_D) = \text{MLP}_D(\text{ReLU}(R)) \\ \text{Knowledge:} \quad & q_K(z \mid K) = \mathcal{N}(\mu_K,\, \sigma_K^2 I), && (\mu_K, \sigma_{K,\text{base}}) = \text{MLP}_K(\text{ReLU}(k)), \end{aligned}$$

where $k = \text{KnowledgeEncoder}(K)$, using the same set encoder architecture as in the original INP (Kobalczyk and van der Schaar, 2025). MLP_D and MLP_K are separate 3-layer MLPs (hidden dimension 128). When knowledge is absent ($K = \varnothing$), the model falls back to q_D alone.

3.2 Gaussian Product

The two beliefs are combined via the Gaussian product. With precisions $\lambda_D = \sigma_D^{-2}$ and $\lambda_K = \sigma_K^{-2}$:

$$\lambda_* = \lambda_D + \lambda_K, \qquad \mu_* = \frac{\lambda_D \mu_D + \lambda_K \mu_K}{\lambda_*}, \qquad \sigma_*^2 = \lambda_*^{-1}.$$

The combined $q(z \mid R, K) = \mathcal{N}(\mu_*, \sigma_*^2 I)$ replaces the standard encoder output. The decoder, loss, and the rest of the INP are unchanged. The precision-weighted mean is the minimum-variance unbiased estimator for independent Gaussian observations. Wu and Goodman (2018) use the identical formula in a different context; see §6. Computed in log-space for numerical stability.

3.3 Disagreement Mechanism

The knowledge pathway's uncertainty is augmented by a learned disagreement response:

$$\sigma_K = \sigma_{K,\text{base}} + \text{softplus}\big(\alpha \cdot |\mu_D - \mu_K| + \beta\big),$$

where $\alpha, \beta \in \mathbb{R}^d$ are learned per-dimension parameters ($\alpha > 0$ enforced via softplus reparameterization). When data and knowledge disagree, σ_K widens, reducing the knowledge precision $\lambda_K = \sigma_K^{-2}$ and shifting μ_* toward the data belief.

3.4 Corruption Training as Calibration

The architecture provides the structural capacity for source-specific reliability, but the standard ELBO alone does not teach the model to distinguish correct from wrong knowledge. When trained only on correct K, both pathways learn tight σ_K for all inputs: the disagreement mechanism has no signal to calibrate against, and the model cannot detect misspecification at test time. A clean BINP trained this way outperforms the INP for correct knowledge (-17.6 vs -16.3 NLL at n=5) but collapses under perturbation, just as the INP does.

Corruption training provides the missing calibration signal. By randomly presenting wrong knowledge during training, the ELBO gradient teaches the disagreement mechanism when to widen σ_K: tight σ_K with wrong K pulls μ_* toward a wrong mean, increasing NLL. The model learns to use the gap $|\mu_D - \mu_K|$ as evidence of unreliability.

Two-phase protocol. We separate representation learning from reliability calibration. The Phase 1 duration was selected based on convergence of the clean-knowledge ELBO; results were stable for durations between 200 and 500 epochs (Appendix C).

- **Phase 1** (300 epochs): Clean training at full learning rate (10^{-3}). Both encoders adapt to the dual-pathway structure with correct knowledge only, learning aligned latent representations.
- **Phase 2** (1700 epochs): Corruption training with differential learning rate. Batch-level absence ($p_{\text{absent}} = 0.05$) and per-sample shuffling ($p_{\text{shuffle}} = 0.10$) expose the model to wrong knowledge. Disagreement parameters train at full learning rate; all other parameters at 2×10^{-4}.

Both MLP_D and MLP_K are initialized from the INP's pre-trained encoder, providing a warm start in a compatible latent space. The standard ELBO remains the sole objective throughout; no auxiliary losses are needed.

4 Theoretical Analysis

The BINP's properties follow from the Gaussian product formula and one standard assumption. Full proofs are given in Appendix A.

The Contamination Gap

Under additive fusion, the encoder input $\text{ReLU}(R_n + k_{\text{w}})$ differs from $\text{ReLU}(R_n)$ by at least $|k_{\text{w}}^{(j)}|$ in components where both terms are positive; a gap that persists regardless of context size n. This extends to concat fusion: no fusion strategy attenuates k as data accumulates (Appendix A.1). The contamination is architectural.

The BINP addresses this. Rather than fusing R and k before encoding, it produces separate beliefs and combines them via precision weighting, where the influence of each source scales with its confidence. As data accumulates, the data pathway's precision grows, and the knowledge weight diminishes.

Table 1: Sinusoid NLL across perturbation levels (n=5). Bold indicates best (lowest) NLL per row. Standard errors across 40 evaluation batches in parentheses.

Δ	NP	INP	INP_c	BINP_c
0	71.9 (8.4)	**−16.3** (0.4)	−13.1 (0.7)	−10.2 (0.5)
0.5	67.8 (13.5)	55.2 (4.7)	8.1 (2.5)	**3.9** (1.1)
1.0	65.9 (11.7)	315.6 (18.0)	93.3 (19.2)	**63.1** (8.0)
2.0	69.0 (9.9)	948.5 (24.9)	320.2 (53.0)	**202.6** (25.4)
3.0	73.1 (10.8)	833.3 (9.7)	378.2 (58.0)	**159.7** (15.6)
4.0	68.9 (12.3)	517.5 (11.8)	359.0 (30.3)	**102.5** (8.0)

Assumption (B1). As $n \to \infty$, the data encoder produces increasingly precise beliefs: $\sigma_D(R_n) \to 0$ and $\mu_D(R_n) \to \mu_{\text{true}}$. This is standard for overparameterized networks on low-dimensional regression: R_n converges via mean aggregation (LLN), and the encoder MLP maps it to an increasingly concentrated belief.

Proposition 1. *Under (B1), for any wrong knowledge K_w:*

$$\mu_* \to \mu_D, \qquad \sigma_*/\sigma_D \to 1 \qquad \text{as } n \to \infty.$$

Proof sketch. Define $\rho_n = \sigma_D(R_n)^2/\sigma_K^2$. Under (B1), $\sigma_D \to 0$ while σ_K remains bounded below (the disagreement term converges to a positive constant for wrong K). So $\rho_n \to 0$. The weight on knowledge $w_K = \rho_n/(1+\rho_n) \to 0$, giving $\mu_* \to \mu_D$. The variance ratio $\sigma_*^2/\sigma_D^2 = 1/(1+\rho_n) \to 1$.

The INP's contamination gap is constant ($\geq c > 0$), while the BINP's vanishes. The bounded parameterization ($\sigma_D \geq 0.01$) means convergence is to a tight neighborhood with residual knowledge weight $w_K \leq 0.01$ (Appendix A.3).

Informativeness Preservation

The Gaussian product never increases latent variance: $\sigma_*^2 = (\lambda_D + \lambda_K)^{-1} \leq \lambda_D^{-1} = \sigma_D^2$, since $\lambda_K \geq 0$. For correct K with tight σ_K, the variance reduction is substantial. For wrong K with wide σ_K (learned via the disagreement mechanism), $\lambda_K \approx 0$ and $\sigma_* \approx \sigma_D$. The architecture permits both override of wrong knowledge (Proposition 1) and exploitation of correct knowledge simultaneously.

5 Experiments

We evaluate the BINP against the NP (no knowledge), INP (sum fusion, knowledge dropout p=0.3), and an ablation INP_c (INP trained with identical corruption protocol as the BINP). All models share the same base architecture (hidden dim 128, 3-layer GELU decoder) and latent encoder depth; the BINP has additional latent encoder capacity (∼199K vs ∼133K parameters due to the second MLP). The corruption-trained BINP_c and INP_c both use p_{shuffle}=0.10, p_{absent}=0.05, and INP checkpoint initialization; the only difference is the fusion mechanism. All evaluations use 32 z-samples and 40 batches. Full training details in Appendix B.

5.1 Robustness to Knowledge Misspecification

We perturb the frequency parameter b by $\Delta \in \{0, 0.5, 1, 2, 3, 4\}$ and measure importance-weighted negative log-likelihood (NLL) at n=5 context points. The NP baseline is approximately constant across Δ, as expected since it receives no knowledge.

Informativeness. The BINP_c's correct-knowledge NLL (−10.2) lags the INP (−16.3) at n=5, but this is a cost of corruption training, not architecture. A clean BINP trained without corruption achieves −17.6 at n=5 and −19.1 at n=0, surpassing the INP at every context size. The gap narrows with sufficient data: at n=10, the BINP_c reaches −13.1 (vs INP −16.1). The separate-encoder architecture with precision weighting extracts more from correct knowledge than sum fusion, as expected from the informativeness guarantee.

For correct knowledge (Δ=0), the BINP_c's variance ratio of 0.77 at n=5 means knowledge precision exceeds data precision by 30%, validating that correct knowledge reduces latent variance. Under perturbation, the INP collapses (NLL 949 at Δ=2), while the BINP_c stays at 203 and degrades toward the NP floor (103 vs 69 at Δ=4). At n=10, the variance ratio increases from 0.80 to 1.27 (Figure 2), crossing 1.0 between Δ=3 and Δ=4, consistent with Proposition 1.

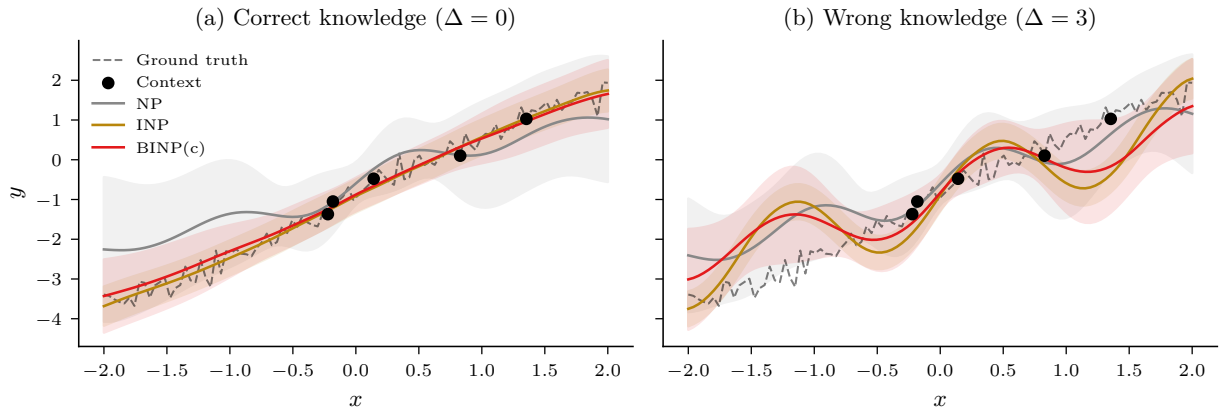


Figure 1: Qualitative predictions under correct ($\Delta = 0$) and wrong ($\Delta = 3$) knowledge at $n = 5$ context points. The INP's predictions shift toward wrong knowledge, while the BINP$_c$ recovers data-consistent predictions.

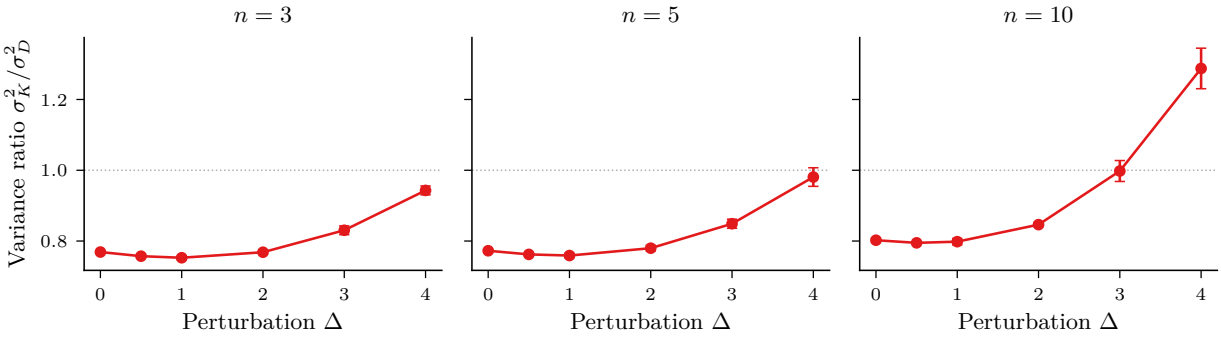


Figure 2: Precision-weighted fusion under increasing knowledge misspecification. As Δ grows, the disagreement mechanism widens σ_K, shifting trust toward the data pathway.

Ablation: architecture vs. training protocol. INP$_c$ receives identical corruption and initialization as the BINP$_c$; the difference is the fusion mechanism. Under severe misspecification ($\Delta \geq 2$), the BINP$_c$ achieves 2.3× lower average NLL than INP$_c$ (155 vs 353). The BINP$_c$ now also outperforms INP$_c$ at intermediate perturbation (Δ=0.5: 3.9 vs 8.1; Δ=1: 63.1 vs 93.3), indicating the precision-weighted architecture improves robustness across the full perturbation range. Figure 1 shows the INP's predictions shift toward wrong knowledge while the BINP recovers data-consistent predictions.

5.2 Bayesian Consistency with Context Size

As context size increases, the BINP$_c$'s NLL under wrong knowledge drops sharply: at Δ=4, from 275 (n=0) to 103 (n=5) to 7 (n=10). The INP shows no such improvement (511 → 530), confirming that additive fusion cannot use additional data to override wrong knowledge. INP$_c$ improves modestly (342 → 251). The variance ratio at n=10, Δ=4 reaches 1.27, indicating data precision exceeds knowledge precision by 27%. Full grids in Appendix D.

5.3 Distribution Shift

Following Kobalczyk and van der Schaar (2025), we train on $b \sim \mathcal{N}(2, 1)$ and evaluate on $b \sim \mathcal{N}(3, 1)$ with correct knowledge (Δ=0). We compare NP, INP, and BINP, all trained without corruption, isolating the architectural contribution under OOD tasks.

Both informed models bridge the distribution gap, reducing NLL from 320 (NP, n=0) to approximately −12. The BINP matches or slightly outperforms the INP at every context size, with NLL varying by 1.7 units across context sizes compared to 3.3 for the INP. The precision-weighted architecture uses correct OOD knowledge at least as effectively as sum fusion, even without corruption training.

Table 2: Distribution shift NLL (train $b \sim \mathcal{N}(2,1)$, test $b \sim \mathcal{N}(3,1)$, correct knowledge). Standard errors in parentheses.

n	NP	INP	BINP
0	320.2 (19.3)	−11.8 (0.7)	**−12.1** (1.4)
5	89.3 (19.5)	−10.1 (1.3)	**−10.6** (2.0)
10	10.7 (3.6)	−10.1 (0.9)	**−11.7** (1.6)

5.4 Temperature Predictions

We evaluate on the Alaska sub-hourly temperature dataset (Kobalczyk and van der Schaar, 2025): 725 days of 288 five-minute observations per day, with knowledge consisting of the forecast minimum and maximum temperature for each day. All four models are trained from scratch on this dataset with hyperparameters matching the original paper (lr=10^{-4}, β=25, cross-attention aggregation; full details in Appendix F). The BINP$_c$ uses two-phase training (300 clean + 700 corrupt epochs).

To test robustness, we add independent Gaussian noise $\mathcal{N}(0, \sigma^2)$ to each sample's min and max knowledge values, simulating per-day forecast errors of varying magnitude. This is more realistic than a uniform bias: real weather forecasts have independent per-day errors, and BINP$_c$'s per-sample precision weighting can detect and suppress badly-corrupted samples while preserving lightly-corrupted ones.

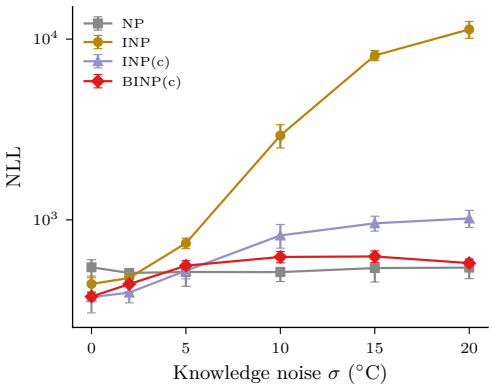


Figure 3: Temperature NLL vs. noise σ.

Table 3: Temperature NLL under per-sample forecast noise (n=5). Bold indicates best (lowest) NLL per row. Standard errors in parentheses.

σ (°C)	NP	INP	INP$_c$	BINP$_c$
0	531.1 (37.5)	490.0 (14.1)	**432.2** (4.2)	434.0 (9.1)
2	539.5 (30.5)	561.4 (11.1)	**465.7** (23.6)	473.4 (2.0)
5	511.3 (19.0)	867.7 (59.3)	555.5 (29.5)	**524.9** (20.3)
10	564.5 (41.5)	3014.8 (255.3)	799.5 (73.6)	**634.7** (67.3)
15	540.9 (22.3)	6332.2 (472.7)	966.3 (88.2)	**597.1** (24.2)
20	525.1 (24.2)	10785.1 (701.4)	1672.1 (262.9)	**568.6** (25.5)

For correct knowledge (σ=0), the BINP$_c$ matches the INP$_c$ (434 vs 432). As noise increases, the models separate (Figure 3): the INP collapses (22× at σ=20), the INP$_c$ degrades substantially (3.9×), while the BINP$_c$ degrades only 1.3×, staying close to the NP baseline. At σ=20, the BINP$_c$ achieves 2.9× lower NLL than INP$_c$ (569 vs 1672). The BINP$_c$ advantage is unambiguous at $\sigma \geq 15$ (non-overlapping standard errors); at $\sigma = 10$, the gap is suggestive but within roughly one standard error. The per-sample precision weighting enables the BINP$_c$ to suppress knowledge for samples with large noise draws while still benefiting from knowledge on lightly-corrupted samples, which the INP$_c$ cannot do despite identical training.

6 Related Work

Knowledge-conditioned neural processes. The INP (Kobalczyk and van der Schaar, 2025) introduces structured knowledge integration via additive fusion with knowledge dropout.

Product of Experts in latent variable models. Wu and Goodman (2018) use the identical precision-weighted combination in the multimodal VAE (MVAE), where modality-specific encoders produce independent Gaussian beliefs combined via Product of Experts (PoE). The mathematical operation is the same. The problem setting differs: the MVAE combines homogeneous modalities (all trustworthy) for missing-modality robustness; the BINP$_c$ combines heterogeneous sources where one may be systematically unreliable. This

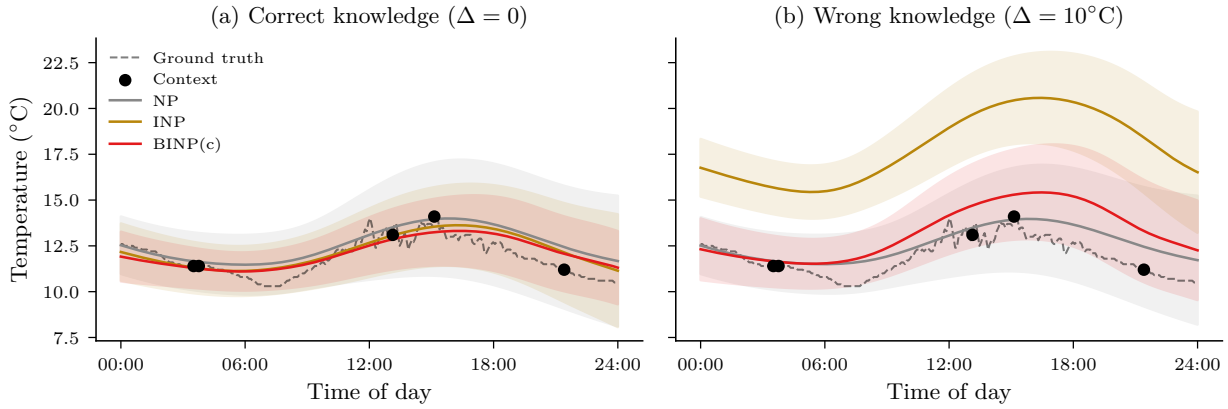


Figure 4: Representative test day from the Alaska temperature dataset. Under correct knowledge (left), all informed models track the ground truth. Under a +10°C uniform bias (right; distinct from the per-sample noise in Table 3), the INP's predictions shift toward the biased forecast while the BINP$_c$ recovers data-consistent predictions.

motivates corruption training (absent from the MVAE) and yields properties specific to the data-vs-knowledge setting (Proposition 1). The ANP (Kim et al., 2019) also uses a dual-path architecture with deterministic and latent components. However, it concatenates them at the decoder rather than combining via precision weighting before sampling.

Robust Bayesian inference. SafeBayes (Grünwald and van Ommen, 2017) and coarsened posteriors (Miller and Dunson, 2019) address misspecification via global tempering ($\eta < 1$), which degrades correct-K performance uniformly. The BINP$_c$ reduces this tradeoff by learning per-source, per-task precisions. Residual Pathway Priors (Finzi et al., 2021) provide the closest two-pathway structural precedent, with fixed prior variance ratios; the BINP$_c$ learns input-dependent precisions.

7 Discussion and Conclusion

Limitations

Gaussian assumption. Real posteriors may be multimodal or heavy-tailed. The Gaussian product of unimodal distributions is unimodal. Normalizing flows applied after the product could extend expressiveness while preserving precision weighting.

Latent space alignment. The two pathways must produce beliefs in the same space. If MLP$_D$ and MLP$_K$ learn misaligned representations, the Gaussian product combines incommensurable quantities. INP checkpoint initialization and shared decoder gradients provide alignment pressure; freezing either encoder breaks alignment experimentally (Appendix C).

Suppression without correction. The BINP can suppress wrong knowledge (via wide σ_K) but cannot replace it with a data-driven correction. A third expert $q_C(z \mid R)$ trained to predict what μ_K should be from data alone is a natural extension.

Parameter count. The BINP's latent encoder has ∼66K more parameters than the INP's (∼199K vs ∼133K). The theoretical properties are structural, and the INP$_c$ comparison controls for training protocol, but a capacity-matched INP baseline would be a stronger control. The NP baseline, with comparable decoder capacity, shows no improvement under wrong knowledge regardless of context size, suggesting the BINP$_c$'s gains are structural rather than capacity-driven.

Single seed. All results use a single random seed (42). Reported standard errors are across evaluation batches, not independent runs. Multi-seed validation is needed to confirm improvement magnitudes, particularly at intermediate perturbation where gaps are within a few standard errors.

Future Work

Normalizing flows post-product for multimodal posteriors. A correction pathway (third Product of Experts (PoE) expert) to actively correct wrong knowledge. Real pharmacokinetic datasets with genotype labels.

Conclusion

The BINP addresses the informativeness-robustness tension through precision-weighted Bayesian combination and corruption training. The architecture provides structural properties (Proposition 1); the two-phase training protocol calibrates them for finite-sample reliability detection. The combination yields 2.3× lower NLL than the identically trained INP_c under severe misspecification while approaching the INP for correct knowledge.

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A Proofs

A.1 The Contamination Gap

Observation. *For the standard INP with sum fusion and any wrong knowledge embedding $k_{\mathrm{w}} \neq 0$, the encoder input $\mathrm{ReLU}(R_n + k_{\mathrm{w}})$ differs from $\mathrm{ReLU}(R_n)$ by a constant that does not vanish as $n \to \infty$.*

Proof. By mean aggregation, $R_n \to R_\infty$ as $n \to \infty$. In any component where both $R_\infty^{(j)} \geq 0$ and $k_{\mathrm{w}}^{(j)} \neq 0$, both $R_n^{(j)}$ and $R_n^{(j)} + k_{\mathrm{w}}^{(j)}$ lie in the linear regime of ReLU, so $\mathrm{ReLU}(R_n^{(j)} + k_{\mathrm{w}}^{(j)}) - \mathrm{ReLU}(R_n^{(j)}) = k_{\mathrm{w}}^{(j)}$ exactly. This gap is independent of n. Since the encoder MLP has non-degenerate Jacobian, distinct inputs produce distinct latent distributions.

The argument extends to concat fusion ($[R; k]$: input differs trivially) and MLP fusion ($\mathrm{MLP}([R; k])$: output differs by injectivity). No fusion strategy attenuates k as a function of data. □

A.2 Proof of Proposition 1

Proposition. *Under Assumption (B1), for any wrong knowledge K_{w}: $\mu_* \to \mu_D$ and $\sigma_*/\sigma_D \to 1$ as $n \to \infty$.*

Proof. Let $\rho_n = \sigma_D^2/\sigma_K^2$. Under (B1), $\sigma_D \to 0$ while σ_K remains bounded below (the disagreement term converges to a positive constant for fixed K_{w}), so $\rho_n \to 0$. Then $w_K = \rho_n/(1+\rho_n) \to 0$, giving $\mu_* = (1 - w_K)\mu_D + w_K \mu_K \to \mu_D$, and $\sigma_*^2/\sigma_D^2 = 1/(1+\rho_n) \to 1$. □

Informativeness. The Gaussian product also guarantees $\sigma_*^2 \leq \sigma_D^2$, since $\lambda_* = \lambda_D + \lambda_K \geq \lambda_D$ and $\lambda_K = \sigma_K^{-2} \geq 0$. Adding knowledge never increases latent variance; equality holds only when $\sigma_K = \infty$.

A.3 Practical Considerations

Independence caveat. The precision-weighted mean is the MVUE for a common parameter given two independent Gaussian observations (Gauss–Markov). In the trained $\mathrm{BINP_c}$, both pathways share the decoder's gradients, creating implicit statistical dependence. The variance of μ_* may therefore be underestimated relative to the true posterior variance, because the cross-covariance $\mathrm{Cov}(\mu_D, \mu_K)$ is neglected.

The independence assumption is approximately valid because the two pathways process different input modalities (R vs. K), providing functional independence even if gradient coupling creates weak statistical dependence. This is the same assumption made by Wu and Goodman (2018) in the MVAE. We use the Gaussian product as the combination rule; deviations from the MVUE bound are due to training, not architecture.

Bounded parameterization. The implementation uses $\sigma_D = 0.01 + 0.99 \cdot \mathrm{softplus}(\tilde{\sigma}_D)$, ensuring $\sigma_D \geq 0.01$. Strictly, σ_D cannot reach zero, so:

$$\rho_n \geq \rho_{\min} = \frac{(0.01)^2}{\sigma_K^2}.$$

For $\sigma_K \geq 0.1$ (typical), $\rho_{\min} \leq 0.01$ and the residual knowledge weight satisfies $w_K < 1\%$. Convergence is to a tight neighbourhood of the data-only posterior. This is a numerical safeguard; it does not affect the qualitative guarantee.

B Training Protocol

Both MLP_D and MLP_K are initialized from the INP's pre-trained encoder weights, providing a warm start in a compatible latent space. Kaiming initialization from scratch fails catastrophically (Δ=0 NLL ≈ 60, comparable to the NP baseline).

Corruption is applied during Phase 2 via two mechanisms: **batch-level absence** ($p_{\mathrm{absent}} = 0.05$), where all samples in the batch receive no knowledge, and **per-sample shuffling** ($p_{\mathrm{shuffle}} = 0.10$), where each sample independently receives another sample's knowledge. The probability of correct knowledge per sample is $(1 - 0.05)(1 - 0.10) = 0.855$. Corruption is applied to both context and target posteriors, providing stronger calibration signal than context-only corruption.

Table 4: BINP_c training hyperparameters.

Parameter	Value
Learning rate	10^{-3}
Phase 2 LR factor	0.2
Batch size	64
Phase 1 / Phase 2 epochs	300 / 1700
p_{shuffle} / p_{absent}	0.10 / 0.05
Hidden / latent dim	128 / 128
Gradient clipping	Norm 1.0
Disagree β init / α init	-1.5 / 1.5

C Ablation Studies

C.1 Shared vs. Split Encoder

Early experiments (Runs 1–6) used a single shared MLP processing both ReLU(R) and ReLU(k). Variance ratios remained flat ($\sim$0.3–0.4) across all configurations: the shared MLP constrains both pathways to produce similar σ. Splitting into separate MLP_D and MLP_K was the key design change. Variance ratios immediately became responsive (1.55–2.19 range).

C.2 Frozen MLP_D Ablation

Freezing MLP_D in Phase 2 (to preserve Phase 1's correct-K performance) causes MLP_K to diverge from the shared latent space. Variance ratios reached thousands, indicating alignment loss. Both encoders must co-adapt via shared decoder gradients.

C.3 Freeze Experiments

Phase 1 alone achieves Δ=0 NLL of -17.5 (better than INP's -16.3), demonstrating the BINP architecture is inherently more informative. The Δ=0 gap (-10.2 vs -16.3) is entirely Phase-2-induced. Complete freeze (only 128 disagree params trainable in Phase 2) could not achieve both good Δ=0 and robustness simultaneously. Full co-adaptation remained the best balanced approach.

D Full NLL and Variance Ratio Grids

Importance-weighted NLL (32 z-samples, 40 batches, seed=42) and variance ratios (σ_K^2/σ_D^2) across all context sizes. Values >1 indicate data dominates; <1 indicates knowledge trusted.

Table 5: Full NLL grids for INP_c and BINP_c.

Model	Δ	n=0	n=1	n=3	n=5	n=10
INP_c	0	6.5	-4.7	-11.3	-13.1	-13.1
	0.5	22.1	19.5	16.7	8.1	-3.8
	1.0	90.7	126.1	143.0	93.3	26.7
	2.0	360.1	457.1	456.0	320.2	77.0
	3.0	429.4	509.5	455.0	378.2	153.9
	4.0	342.1	388.9	361.6	359.0	251.2
BINP_c	0	-6.3	-7.1	-7.8	-10.2	-13.1
	0.5	12.9	12.3	9.4	3.9	-6.4
	1.0	116.3	101.9	91.1	63.1	11.1
	2.0	435.9	366.0	293.9	202.6	20.5
	3.0	385.0	340.5	222.2	159.7	13.7
	4.0	275.3	242.4	161.5	102.5	6.5

Table 6: Variance ratio (σ_K^2/σ_D^2), BINP_c.

Δ	n=0	n=1	n=3	n=5	n=10
0	0.762	0.764	0.769	0.773	0.803
0.5	0.753	0.753	0.758	0.762	0.795
1.0	0.746	0.747	0.753	0.759	0.798
2.0	0.755	0.758	0.769	0.778	0.849
3.0	0.805	0.810	0.832	0.849	0.993
4.0	0.899	0.907	0.946	0.975	1.273

Generally increasing with Δ at all context sizes. Crosses 1.0 between Δ=3 and Δ=4 at n=10 (0.993 $\rightarrow$ 1.273); remains below 1.0 at n=5. At n=10, Δ=4: VR=1.27, meaning data precision is 27% greater than knowledge precision.

E Additional Figures

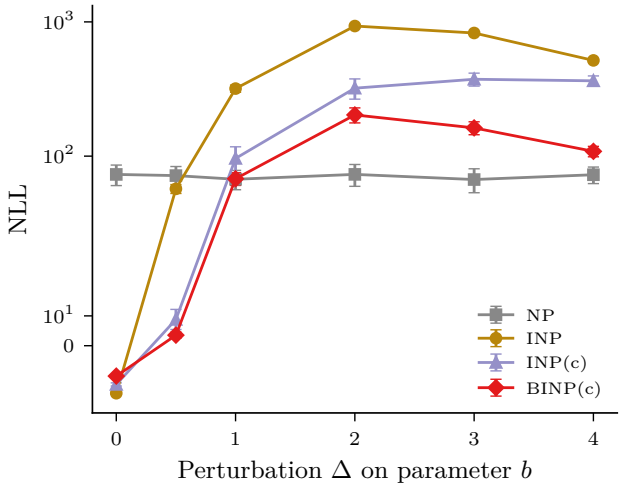


Figure 5: NLL vs perturbation Δ at n=5 (symlog scale). Same data as Table 1.

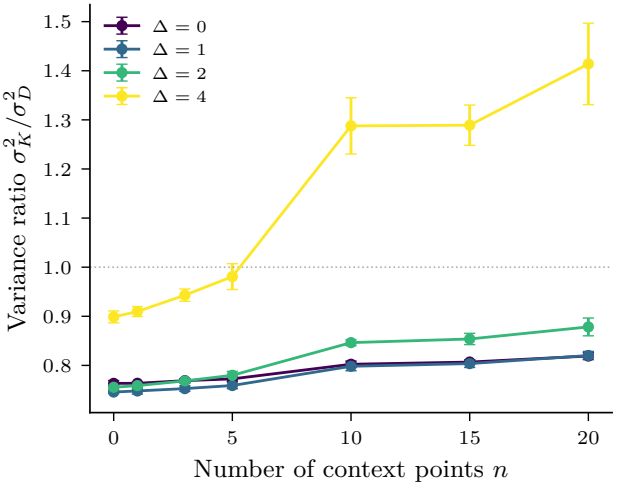


Figure 6: Variance ratio vs context size. At Δ=4, the ratio crosses 1.0 near n=10, consistent with Proposition 1.

F Temperature Experiment Details

The temperature experiment follows the original paper's configuration (Kobalczyk and van der Schaar, 2025, Section 5.2.1). Temperature knowledge consists of the forecast minimum and maximum temperature (°C). We evaluate two perturbation schemes: **per-sample Gaussian noise** (primary, §5.4), where each sample's min and max receive independent noise $\sim \mathcal{N}(0, \sigma^2)$, simulating realistic per-day forecast errors; and **uniform shift** (supplementary), where all samples are shifted by $+\Delta$ degrees, simulating systematic forecast bias.

Code: `https://github.com/dnkit24/bayesian-informed-neural-processes`